Problem Chosen	2020	Team Control Number
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	Summary Sheet	

Build for the Water: Exposure, Moisture and Sandcastle Survival

Summary

The longest-lasting foundation depends on wave direction, rain, moisture evolution and useful height. We couple projected wave/rain exposure to a conserved water balance and compare equal 30 L foundations by the time until half their sand remains.

Two analytic limits give checkable geometric answers. Full contact and fixed moisture yield a top/base radius ratio $(\sqrt{3} - 1)/2 = 0.366$ when the rain/wave ratio is at most 0.386; above that threshold, the optimum reaches a circular boundary with a different top ratio. A broader Brunn–Minkowski bound proves that a circular cone minimizes rain projection over all nonnegative slope-limited height graphs, and attains a sharp rain-only lifetime bound at fixed resistance. Neither result proves a universal optimum under changing moisture and waves.

A synthetic tide scenario finds a curved dry oval lasting 15.62 h versus 9.00 h for the circular cone. Its height is only 1.94 cm; adding a 10 cm useful-height condition selects the cone. Extending the profile family materially improves the dry candidate, demonstrating that a finite shape comparison cannot establish global optimality. A 30 degree direction shift removes its dry advantage, and equal center distance sharply reduces it.

Rain reverses the ranking: at 5 mm/h the dry profile lasts 1.08 h versus 3.90 h for the rain design; at 20 mm/h, 0.27 h versus 1.40 h. Under identical forcing, fixing resistance at its peak extends the 5 mm/h cone lifetime to 20.23 h. The difference isolates moisture feedback within the equations; these hours are uncalibrated scenarios.

The nominal static mixture optimum is 1% liquid by compacted bulk volume. Starting moisture is different: on the cone at low tide, two unmeasured response tails sharing that peak give near-best grid spans of 3.78–5.57% and 33.83–36.22%, using a 0.01 h lifetime tolerance. These are numerical sample ranges, not confidence intervals or beach recipes. Their disagreement supports local strength/drying tests. Volume and density-dependent mass ratios are distinguished.

All 26 named author checks pass, including independent geometric integration, analytic bounds and direct-volume RK4. We recommend compact circular supports when rain, useful height or uncertain direction matters, with deliberate siting, tide timing, packing and water management. A two-page *Fun in the Sun* article translates the conditional results.

Keywords: projected exposure; slope-limited geometry; wet sand; tides; rain.

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1 A useful foundation needs a precise survival criterion

A mound that retains half its sand may have lost its doorway, and a surviving tower may stand on a badly scoured base. We therefore separate a measurable foundation lifetime from aesthetic survival. For a foundation of initial bulk volume V_0, define

$$T_q = \inf\{t \geq 0 : V(t) \leq qV_0\}, \qquad q = 0.50. \tag{1}$$

We also report $q = 0.25$ and 0.75. The foundation is compared before any superstructure is added. A heavy tower would require a separate load-bearing check. Our geometric family remains below a conservative friction-supported side gradient; this prevents the comparison from rewarding a tall, structurally implausible wall merely because it has little horizontal area.

The question fixes the beach, amount and type of sand, water proportion, and approximate distance from the sea. It does not fix the tidal history, direction of waves, available footprint, minimum useful platform height, or the meaning of "last". Those omissions matter. Directed wave erosion can favor a narrow, long mound; rain acts over its entire footprint; changes in moisture can outweigh either geometric effect. We retain these alternatives explicitly. A toe-controlled comparison gives each foundation the same initial seaward-edge distance. A second comparison places the centers equally far inland. The first isolates geometry after equal wave arrival; the second accounts for the fact that a longer footprint reaches farther seaward.

Our primary answer is a conditional design rule: use a compact, gently sloping circular mound when rain or changing wave directions matter; a flatter oval foundation can improve sand retention when waves approach consistently along its long axis and its toe is held at the same distance. There is no supported universal best solid from the information in the statement. We establish an exact optimum in a tractable limiting family, compute a richer time-dependent comparison, and expose the conditions under which the limiting optimum fails.

2 Mechanisms, assumptions and information limits

Small quantities of water create capillary bridges between grains; full immersion destroys ordinary liquid–air bridges. Wet granular material also exhibits compaction, friction, and a dependence on its liquid state. The review by Mitarai and Nori reports a dry sand surface angle near 35° and explains why wet sand can sustain steeper shapes [2]. We use $m_{\max} = \tan 35^\circ = 0.7002$ as a conservative design bound, rather than asserting that steeper wet shapes cannot exist.

Pakpour and colleagues measured the stiffness and buckling of compacted sand columns, finding a modulus maximum near a reported liquid volume fraction of 1% [1]. That is evidence about a particular static material and preparation, not a calibrated sea-wave erosion law. We use the reported optimum to motivate a nominal response curve. The paper's wording does not resolve a precise bulk/solid denominator transfer for our mixture variable; that uncertainty is retained when discussing recipes.

The common assumptions have direct consequences:

1. **Homogeneous material and equal packing.** Common material coefficients allow geometry to be compared at equal sand volume and initial liquid fraction. Different grains

or compaction would require new coefficients.

2. **Exported erosion.** Detached sand leaves the foundation rather than immediately redepositing. The volume equation is inappropriate where returning swash rebuilds the mound.

3. **Homothetic shape reduction.** Erosion reduces all dimensions by a common factor. This closes the area–volume relation but omits overhangs, localized toe scour and gullies. Such events can cause earlier structural failure.

4. **Averaged wave forcing.** A time-varying effective water depth represents wave contact and tide. It is a scenario input, not an observation or a full hydrodynamic solution.

5. **Lumped water content.** Water mixes through an active bulk volume and departing sand carries the same fraction. Surface drying fronts and preferential infiltration could violate this closure.

6. **Vertical rainfall and a separate erosion efficiency.** Rain changes both material moisture and loss of sand. A slope-dependent runoff law is examined as a limitation, not inserted without data.

The numerical domain $0 \leq \alpha \leq 0.75$, $1 \leq \lambda \leq 3$, $0.25 \leq m \leq 0.7002$ describes a practical set of elliptical height profiles, including frusta and curved mounds, with a profile exponent $1 \leq p \leq 4$. These are illustrative construction choices, not restrictions in the original task. A separate $H \geq 0.10$ m scenario represents a builder who needs a useful raised platform; it is not silently substituted for the original comparison.

Table 1. Inputs and their evidence status

Input	Nominal value	Basis and consequence
V_0	0.030 m^3	Common illustrative bulk volume; no supplied measurement.
θ_*	0.010	Transferred material-response scenario motivated by [1]; denominator and erosion transfer uncalibrated.
$m_{\max}$	0.7002	Conservative dry-angle design, motivated by [2]; excludes no broader wet geometry by proof.
v_w	0.015 m h^{-1}	Synthetic recession scale; no measured beach coefficient.
k_r	0.20	Synthetic sand recession per rain depth; not a soil coefficient transferred to sand.
P	0, 5, 20 mm h^{-1}	Steady illustrative rain scenarios, not probabilities.
e, i_w	0.1, 0.2 mm h^{-1}	Synthetic evaporation and wave infiltration speeds.
k_d, θ_f	0.30 h^{-1}, 0.005	Synthetic drainage rate and residual fraction.
$D(t)$	Mean 0.04 m; amplitude 0.12 m	Effective contact-depth scenario with period 12 h 25 min, a representative semidiurnal period [3].

No absolute lifetime in hours below is a field forecast. The common coefficients can be measured locally; the model then becomes a calibrated comparison. Without them, the numerical examples illustrate decisions and identify measurements that can change them.

3 Geometry turns forcing into exposure

Let a and b be the long and short base semiaxes, H the height, and α the ratio of top to base semiaxes. An elliptical frustum has horizontal section

$$\frac{x^2}{a^2 f(z)^2} + \frac{y^2}{b^2 f(z)^2} \le 1, \quad f(z) = 1 - (1-\alpha)\frac{z}{H}, \quad 0 \le z \le H. \tag{2}$$

Thus $\alpha = 0$ is a cone, while $\alpha > 0$ supplies a top platform. Set $\lambda = a/b \ge 1$ and $m = H/[b(1-\alpha)]$, the maximum side gradient. Slicing horizontally gives

$$V = \frac{\pi a b H}{3} F(\alpha), \qquad F(\alpha) = 1 + \alpha + \alpha^2. \tag{3}$$

For a wave moving along the long axis and wetting the bottom $z_w = \min\{H, \max(D, 0)\}$, the frontal projected area is

$$A_w = 2b \int_0^{z_w} f(z)\, dz = 2b \left[z_w - \frac{(1-\alpha) z_w^2}{2H} \right]. \tag{4}$$

To test whether straight sides artificially select the answer, we also use

$$f_p(z) = [1 - (1 - \alpha^p)z/H]^{1/p}, \quad 1 \le p \le 4. \tag{5}$$

Here $p = 1$ recovers the frustum and $p = 2, \alpha = 0$ gives a paraboloid. The maximum side gradient is $m = pH/[b(1 - \alpha^p)]$, attained at the short-axis base edge. Direct integration gives

$$V = \pi abHI_p, \qquad I_p = \frac{p}{p+2}\frac{1-\alpha^{p+2}}{1-\alpha^p}, \tag{6}$$

$$A_w = \frac{2bHp}{(1-\alpha^p)(p+1)}\left[1 - (1 - (1-\alpha^p)z_w/H)^{1+1/p}\right]. \tag{7}$$

All candidate profiles obey the same gradient and material limits. For $p = 1$, full contact gives $bH(1 + \alpha)$. The projected area seen by vertical rain is $A_r = \pi ab$. These projections represent flux-driven detachment. They are not the wetted surface area of a 3D fluid solution.

For a direction offset φ from the long axis, replace b in (4) by

$$b_\varphi = \sqrt{b^2 \cos^2\varphi + a^2 \sin^2\varphi}. \tag{8}$$

An oval therefore has a small wave front only in the intended orientation. Turning it through 90° multiplies that breadth by λ. A circular base is orientation independent. Figure 1 shows the two exposure directions. The cross-wave section is schematic. The dot-in-circle shows wave motion along the long axis toward the viewer, normal to this $2b$ section. Vertical rain reaches the horizontal footprint.

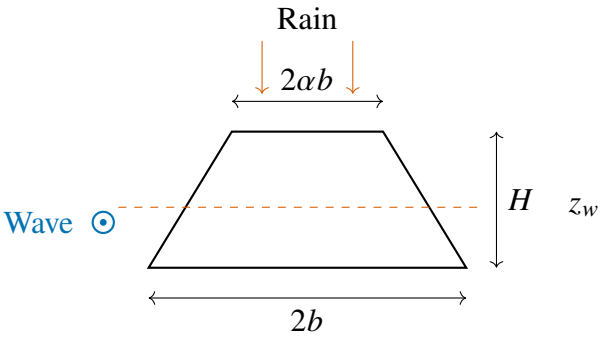


Figure 1. Directional wave exposure

The time-dependent foundation uses a scale $\ell = (V/V_0)^{1/3}$, so $a = a_0\ell$, $b = b_0\ell$, and $H = H_0\ell$. The toe-controlled depth is

$$D(t) = 0.04 + 0.12\sin(2\pi t/12.4167 - \pi/2) \quad \text{m}. \tag{9}$$

It begins below the base. In the center-controlled alternative, we add $s(a - 0.50)$ with an illustrative beach slope $s = 0.04$, because a longer footprint has a more seaward toe. This is a geometric first-order sensitivity, not an exact planar-beach inundation calculation.

4 A conserved erosion and water model

Define $\theta = W/V$, where W is the volume of pore water and V the current compacted bulk volume. Water replaces pore air; adding one liter need not expand the bulk by one liter. The nominal dimensionless resistance is

$$C(\theta) = \varepsilon + (1-\varepsilon)x\exp(1-x), \quad x = \theta/\theta_*, \quad \varepsilon = 0.10. \tag{10}$$

The positive floor describes residual frictional resistance in this reduced erosion model. The response rises and then falls as wetting increases. Its precise curve is phenomenological and does not reproduce the full pendular–funicular–capillary sequence. In particular, extrapolating it to very high initial water fractions is a diagnostic scenario rather than a reliable material recipe.

The bulk sand loss is

$$\dot{V} = -\frac{v_w A_w + k_r P A_r}{C(\theta)}. \tag{11}$$

The speeds v_w and $k_r P$ have units m h^{-1}; their product with an area gives m^3 h^{-1}. All terms are nonnegative, so the modeled foundation never gains sand. Changing wave stress, duty cycle, grain shape or packing can change these speeds; their commonality is an assumption used for the geometry comparison.

Rainfall, wave infiltration, evaporation and drainage change W. If eroded bulk carries its local water fraction, the water balance is

$$\dot{W} = PA_r + i_w A_w - eA_r - k_d V(\theta - \theta_f)_+ + \theta\dot{V}. \tag{12}$$

Differentiating $\theta = W/V$ cancels the departing-water term, leaving

$$\dot{\theta} = \frac{PA_r + i_w A_w - eA_r}{V} - k_d(\theta - \theta_f)_+. \tag{13}$$

At $\theta = 0$, outward water flux is limited to keep the state nonnegative. At the illustrative porosity 0.40, further inflow is treated as overflow. These are reflecting storage boundaries; omitted runoff erosion is a model limit. The state variable ℓ obeys $\dot{\ell} = \dot{V}/(3V_0\ell^2)$.

This coupling explains why a design with a low exposed wave front can still fail in rain. A thin foundation has large A_r/V, so the same rain depth changes its liquid fraction quickly. Rain also removes sand directly through the second term in (11). A model that adds rain to wave forcing but freezes moisture misses this second route to a changed ranking.

5 An exact shape optimum in a limiting case

An analytic case makes the numerical model testable. Suppose the entire side is continuously wet, moisture is fixed, the wave direction is known, and no upper bound is imposed on ellipse elongation. Maximizing T_q is equivalent to minimizing

$$J = A_w + \rho A_r, \qquad \rho = \frac{k_r P}{v_w} > 0. \tag{14}$$

At fixed H and α, volume fixes ab. Reducing b decreases the wave front without changing rain area. The active slope constraint therefore gives $b = H/[m_{\max}(1-\alpha)]$ whenever the resulting

$a \geq b$ is admissible. Substitution into the objective yields

$$J(H, \alpha) = c(\alpha)H^2 + \frac{3\rho V_0}{HF(\alpha)}, \quad c(\alpha) = \frac{1+\alpha}{m_{\max}(1-\alpha)}. \tag{15}$$

For fixed α, $J'' = 2c + 6\rho V_0/[FH^3] > 0$. There is exactly one minimum,

$$H_*^3 = \frac{3\rho V_0}{2cF}. \tag{16}$$

After minimizing in H, the remaining objective is proportional to $[c/F^2]^{1/3}$. Direct differentiation gives

$$\frac{d}{d\alpha} \log \frac{c}{F^2} = \frac{2\alpha(2\alpha^2 + 2\alpha - 1)}{(1-\alpha^2)F}. \tag{17}$$

It is negative for small positive α, zero at

$$\alpha_* = \frac{\sqrt{3}-1}{2} = 0.3660254, \tag{18}$$

and positive thereafter. The endpoint $\alpha \to 1$ has diverging objective. This is a relaxed stationary shape until its ellipse is checked. At that height the optimal aspect is

$$\lambda_*(\alpha) = \max\left\{1, \frac{2m_{\max}(1-\alpha^2)}{\pi\rho}\right\}. \tag{19}$$

To see this directly, at fixed α the volume and active gradient give $J = K(\alpha)[m_{\max}(1-\alpha^2)\lambda^{-2/3} + \pi\rho\lambda^{1/3}]$, with $K(\alpha) > 0$. Differentiation in λ gives the expression above. Thus the constant ratio in (18) is feasible and globally optimal in the declared frustum family only when

$$0 < \rho \leq \rho_c = \frac{2m_{\max}(1-\alpha_*^2)}{\pi} = 0.386045. \tag{20}$$

For $\rho > \rho_c$, the global optimum is on the circular boundary:

$$\lambda = 1, \qquad \alpha_c = \frac{m_{\max}}{m_{\max} + \pi\rho}, \quad R^3 = \frac{3V_0}{\pi m_{\max}(1-\alpha_c^3)}, \quad H = m_{\max}R(1-\alpha_c). \tag{21}$$

Indeed, the circular objective is proportional to $[m_{\max}(1-\alpha^2) + \pi\rho]/[m_{\max}(1-\alpha^3)]^{2/3}$. Its derivative changes from negative to positive at α_c. For a fixed $\rho < 2m_{\max}/\pi$, the transition is $\alpha_t = \sqrt{1 - \pi\rho/(2m_{\max})}$: the elliptical branch applies for $\alpha \leq \alpha_t$ and the circular branch thereafter. When $\rho \leq \rho_c$, $\alpha_* \leq \alpha_t$ and the circular stationary point lies before its admissible interval, so the circular part only increases after the transition. When $\rho > \rho_c$, the elliptical part decreases up to the transition and the circular part continues decreasing until α_c, then increases. If $\rho \geq 2m_{\max}/\pi$, the entire admissible branch is circular. The two optimum branches agree at ρ_c. These derivative signs establish the piecewise global optimum for all $\rho > 0$ in this fixed-moisture, full-contact frustum class. For $\rho \leq \rho_c$, the top/base area ratio is 13.40%; it is not a constant for every positive rain/wave ratio.

At $\rho = 0.50$, the feasible circular optimum is $\alpha = 0.308325$ and $J = 0.266986018$ m^2. Forcing the unconstrained constant ratio onto a circular foundation yields a strictly larger objective. This comparison illustrates why the aspect constraint must be enforced before declaring an optimum.

At $\rho = 0.10$, $V_0 = 30$ L and $m_{\max} = 0.7002$, the calculation gives $H_* = 0.0992$ m and $a/b = 3.86$. Its elongation lies outside the numerical $\lambda \le 3$ construction domain, illustrating why a limiting optimum cannot be silently substituted for a constrained solution. The circular boundary is handled by (21); an additional construction-footprint bound would require another constraint branch.

Under constant coefficients, projected areas scale as ℓ^2. Integrating (11) gives

$$T_q = \frac{3V_0C}{v_w J_0}(1 - q^{1/3}). \tag{22}$$

This exact lifetime is independent evidence for the numerical erosion integrator in its appropriate limit.

Two broader-domain observations prevent overclaiming. With $P = 0$ and a fixed directed wave front, choosing $H \to 0$, $b \propto H$ and $a \propto H^{-2}$ preserves volume while $A_w \propto H^2 \to 0$. There is then no finite best shape in this model without a footprint or functional-height constraint. Conversely, for a fixed cross-sectional profile, volume and height fix ab. The worst directional breadth is $\max(a, b) \ge \sqrt{ab}$, with equality only for a circle. A circular footprint is therefore minimax against unknown horizontal wave directions within that profile class. Neither argument proves a global optimum among all possible solids, but both explain which missing specifications matter.

6 A sharp rain bound beyond the chosen shape family

A broader result supports the circular rain design without relying on a sampled list of solids. Consider any nonnegative, compactly supported height function $h : \mathbb{R}^2 \to \mathbb{R}$ satisfying

$$|h(x) - h(y)| \le m_{\max}|x - y| \quad \text{for all } x, y. \tag{23}$$

This includes irregular footprints, curved slopes and multiple peaks, but excludes overhangs and cohesive vertical walls. Let $E_z = \{x : h(x) \ge z\}$ and $A_0 = |\{h > 0\}|$. For $0 < z < H_{\max} = \max h$, the level set E_z is nonempty and compact. If $x \in E_z$ and $|u| < z/m_{\max}$, the gradient bound implies $h(x + u) > 0$. Therefore $E_z + B_r \subset \{h > 0\}$ for every $r < z/m_{\max}$, where B_r is a disk. At a point of maximum height, the same argument places an open disk of radius $H_{\max}/m_{\max}$ inside $\{h > 0\}$; hence $H_{\max} \le m_{\max}\sqrt{A_0/\pi}$. Levels above this height are empty.

The planar Brunn–Minkowski inequality [5] gives $\sqrt{A_0} \ge \sqrt{|E_z|} + \sqrt{\pi}r$. Letting $r \uparrow z/m_{\max}$ and integrating horizontal layer areas yields

$$V = \int_0^\infty |E_z|\, dz \le \int_0^{m_{\max}\sqrt{A_0/\pi}} \left(\sqrt{A_0} - \frac{\sqrt{\pi}z}{m_{\max}}\right)^2 dz = \frac{m_{\max}A_0^{3/2}}{3\sqrt{\pi}}. \tag{24}$$

Hence the minimum possible horizontal rain projection is

$$A_{\min}(V) = \left(\frac{3\sqrt{\pi}V}{m_{\max}}\right)^{2/3}. \tag{25}$$

The circular cone $h(x) = m_{\max}(R - |x|)_+$ attains every inequality: its level sets are nested disks with linearly decreasing radii. This is an attained global bound over the stated height-graph class, not a grid optimum.

For uniform rain-only projected erosion and fixed common resistance C, any admissible evolving foundation obeys $-\dot{V} \geq k_r P A_{\min}(V)/C$. Integrating gives the sharp upper bound

$$T_q \leq \frac{3CV_0^{1/3}(1-q^{1/3})}{k_r P(3\sqrt{\pi}/m_{\max})^{2/3}}, \tag{26}$$

attained by the selfsimilar cone in that erosion closure. At 30 L, its footprint is 0.3730 m^2, radius 0.3446 m and height 24.13 cm. A square pyramid with the same maximum side gradient needs 0.4043 m^2, or 8.39% more projected rain area.

The theorem supplies a limiting reason to prefer a circular cone in rain. It does not prove a global optimum for changing moisture, waves, runoff or unrestricted cohesive architecture. At 5 mm/h and $C = 1$, the rain-only bound is 49.78 h. The tide-coupled model introduces both wave loss and a changing resistance; their effects are separated in Table 5.

7 Dry-weather results and a usable platform

We searched the original frustum domain and then the broader $p \in [1, 4]$ profile domain using differential evolution with a fixed seed and bounded polishing. Every domain corner was also evaluated. The iteration limit was reached; these are searched candidates, not certified global dynamic solutions. Curved comparators are included to test whether the answer follows from the chosen straight-sided family.

Table 2. Equal-volume dry comparisons

Foundation	Height (cm)	p	a/b	α	$T_{0.5}$ (h)
Expanded oval	1.94	4.0	3.00	0.750	15.62
Linear frustum oval	2.53	1.0	3.00	0.750	9.56
Circular cone	24.13	1.0	1.00	0.000	9.00
Circular frustum	15.55	1.0	1.00	0.366	8.58
Elongated cone	16.73	1.0	3.00	0.000	7.61
Flat frustum	5.23	1.0	2.00	0.600	6.94
Paraboloid	13.28	2.0	1.00	0.000	7.42
Fourth power mound	7.60	4.0	1.00	0.000	6.50

In the straight-sided family, the dry oval lasts 9.56 conditional hours. Allowing the fourth-power curved profile finds a flatter oval lasting 15.62 h, 63.39% longer. It lies at all four construction boundaries: $\alpha = 0.75$, $\lambda = 3$, $m = 0.25$ and $p = 4$. Its semiaxes are 1.359 m and 0.453 m, but its height is only 1.94 cm. Increasing the allowed footprint or profile exponent may improve it further; its position on a construction boundary is not evidence of a universal best solid.

This expanded dry candidate spreads 30 L over 1.934 m^2 and faces aligned waves with a full-contact projection of 0.0156 m^2. It outlasts the circular cone's 9.00 h by 73.51% under the equal-toe, known-direction comparison. However, the cone is 24.13 cm high and occupies a much smaller footprint. Requiring an initial height of at least 10 cm in the same expanded search selects the circular cone, with 9.00 h of modeled lifetime.

The thin candidate is the answer to a sand-retention criterion under a specific forcing and construction domain. It is not presented as the best support for every castle. Its very low height, wider footprint, directional dependence and omitted toe scour matter to a builder. We recommend the compact circular design when a raised support or reliable performance across changing directions is needed. A builder who values a surviving low sand platform and knows the direction of swash can deliberately select the oval, accepting these tradeoffs. Figure 2 compares steady rain while keeping initial liquid fraction 0.01, volume 30 L and geometry common. All curves stop at half volume.

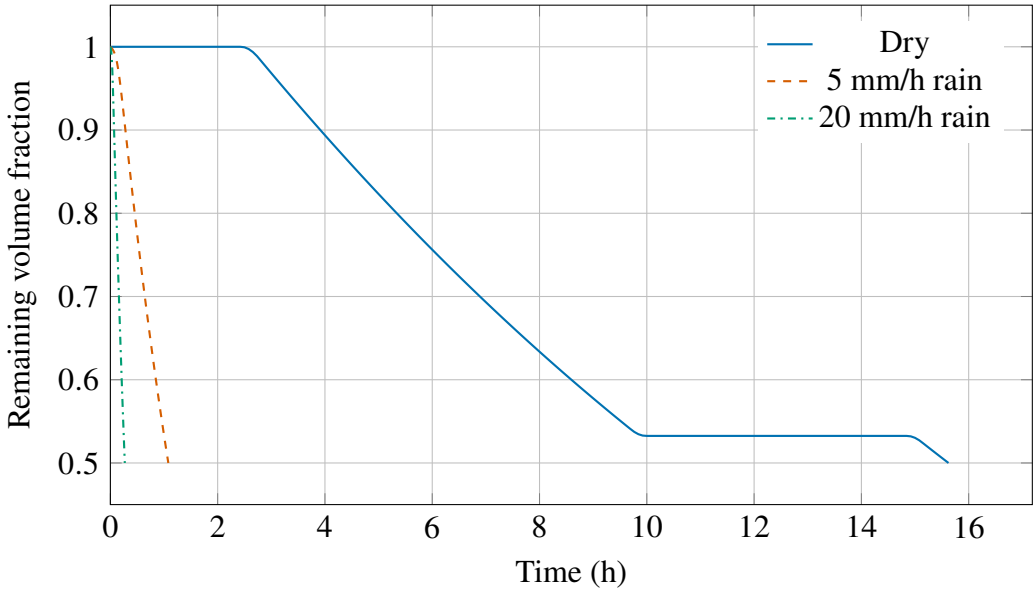


Figure 2. Survival of the dry profile under steady rain

8 Water proportion: static strength versus initial storage

For fixed moisture, differentiating $x\exp(1-x)$ gives $\exp(1-x)(1-x)$, so (10) has its unique maximum at $\theta = \theta_* = 0.010$. This is the model's optimal proportion under constant water content. It corresponds to one liter of pore water per 100 L of final bulk foundation volume, including water already present in the sand. Because that water occupies existing pores, this is not a prescription to combine 99 L of loose sand with one liter of water.

Mass and volume ratios differ. If the *wet bulk* density is 1600 kg m^{-3}, one cubic meter at this fraction contains 10 kg of water in 1600 kg of total mass: 0.625% water by total mass, or approximately 159:1 dry-sand-to-water mass. This conversion is conditional on that density and our denominator. The source experiment reports a liquid volume fraction near 1%, but it does not establish this exact recipe for every beach.

Initial moisture is a different optimization. For each fixed geometry and response law, we searched 401 initial fractions from 0.001 to 0.399, refined every sampled local-maximum neighborhood, and compared both endpoints. The grid spacing is 0.0995 percentage points. To avoid assigning false precision to a flat peak, we report samples satisfying $T_{0.5}(\theta_i) \geq T_{\mathrm{ref}} - 0.01$ h, where T_{ref} is the reported refined candidate lifetime. For the expanded dry profile, 47 samples span 17.41–21.99% and remain within 0.01 h of the 19.48 h reference. The additional water

is a temporary reservoir during the initial dry period, not persistent extra strength. This is a numerical near-best sample set, not a confidence interval, a guarantee for every intervening fraction, or a beach recipe. The search does not certify a joint shape–mixture optimum. Figure 3 joins the 401 simulated initial fractions on the expanded dry profile. The storage optimum differs from the static resistance maximum near 1%; abrupt changes can occur when survival extends to a later tide.

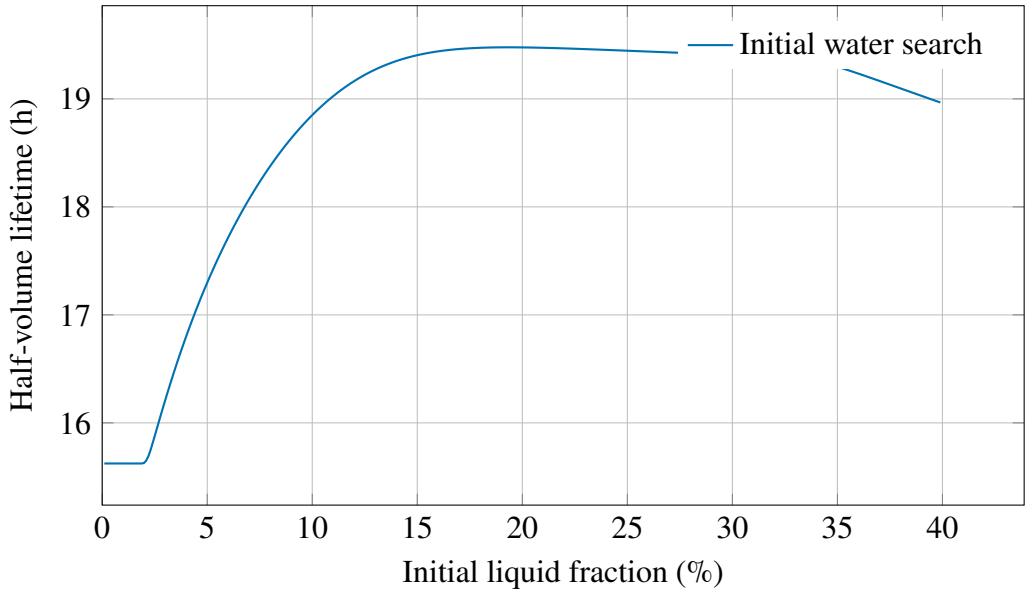


Figure 3. Survival with initial water storage

We attacked the material extrapolation with a second response law that shares the same nominal optimum and loses its cohesive component at saturation. With $n = 0.40$ and $a_\theta = \theta_* n/(n - 2\theta_*)$, let

$$F_R(\theta) = \frac{\theta(n-\theta)/(\theta + a_\theta)^2}{\theta_*(n-\theta_*)/(\theta_* + a_\theta)^2}, \qquad 0 \le \theta \le n. \tag{27}$$

Its derivative has the sign of $na_\theta - \theta(n + 2a_\theta)$, so its unique peak is exactly θ_*. Both laws agree about the location of maximum current resistance, yet their tails differ. Neither tail is fitted to beach erosion.

Table 3. Near-best initial-water grid samples

Shape	Tail	Tide	Grid span (%)	Samples	T_{ref} (h)
Oval	Exp.	Low	17.41–21.99	47	19.48
Cone	Exp.	Low	3.78–5.57	19	16.93
Cone	Exp.	High	5.08–5.27	3	12.76
Cone	Rational	Low	33.83–36.22	25	18.61
Cone	Rational	High	10.55–11.74	13	22.04

Table 3 gives the near-best grid samples on both geometries. On the circular cone at low tide, the exponential tail gives a span of 3.78–5.57%, while the rational tail gives 33.83–36.22%. The spans are widely separated despite the identical static optimum. Starting at high tide removes the initial no-wave waiting period and changes the selected fractions again. Thus a single

measured peak location cannot identify the lifetime-optimal starting recipe. The listed endpoints are rounded grid values; the tolerance concerns calculated lifetime, not measurement uncertainty.

The supplied problem cannot establish a universal mixture ratio. Our conditional answer is near the nominal 1% bulk liquid fraction when moisture is held constant, or the locally calibrated initial-water optimum when drying and wave-arrival timing are known. The large storage optima in Table 3 and Figure 3 are model diagnostics, not recommended beach recipes. They depend on an unmeasured curve over much of the pore-water range. A practical builder should select and maintain a favorable measured material range rather than copy a high initial-water value from this simulation.

A simple calibration can make the advice usable. Take equal compacted samples of the actual sand at several measured water masses, measure their bulk volumes and how much sand is lost under a repeatable shallow wash, and select the broad best region rather than a noisy single maximum. Then observe drying over the expected waiting period. This separates current stiffness from the moisture at first wave contact and directly measures the erosion resistance needed by (11).

9 Rain changes both survival time and preferred shape

In the coupled model, rain is supplied as water depth per time, converted to volume by A_r, and also produces direct sand loss through k_r. The distinction matters even before the tide reaches the foundation: a wide thin oval receives more total rain per retained liter than a taller compact mound.

Table 4. Retaining or changing the dry design in rain

Rain (mm/h)	Dry oval(h)	Rain search(h)	Height (cm)	a/b
5	1.08	3.90	24.13	1.00
20	0.27	1.40	24.13	1.00

At 5 mm h^{-1}, the expanded dry design's half-volume lifetime falls from 15.62 h to 1.08 h. The rain search selects the circular cone with $T_{0.5} = 3.90$ h, 261.23% longer than keeping the dry design. At 20 mm h^{-1}, the respective lifetimes are 0.27 h and 1.40 h, a 420.09% gain. Thus the dry-weather candidate does *not* remain best in these rain scenarios. Both the new curved-profile search and the earlier straight-sided search select the same compact rain shape.

This reversal should not be extrapolated to every rainfall. The analytic full-contact result instead predicts $H_* \propto \rho^{1/3}$ as direct rain erosion grows, until constraints are reached. The coupled examples add changing moisture, partial inundation and a construction domain. Their near-conical optimum is consistent with reducing the large rain-collection area of the thin oval; it does not invalidate the piecewise limiting result under its different assumptions.

To isolate the resistance channel, we hold $C = 1$ while retaining the same geometries, initial liquid fraction, waves, tide phase, rain and toe-distance convention. Rain still removes grains directly, and the water balance remains active; only its feedback into erosion is suppressed.

Table 5. Geometry with and without moisture feedback

	Thin oval		Circular cone	
Rain (mm/h)	$C(\theta)$ (h)	$C = 1$ (h)	$C(\theta)$ (h)	$C = 1$ (h)
0	15.62	> 48	9.00	40.29
5	1.08	8.84	3.90	20.23
20	0.27	2.40	1.40	8.63

Same aligned waves, low starting tide, equal toe distance and $\theta_0 = 0.01$. The fixed-C dry oval is censored at 48 h.

At 5 mm/h, the fixed-resistance cone lasts 20.23 h, compared with 3.90 h when resistance follows moisture. The oval changes from 8.84 h to 1.08 h. Geometry still matters with C fixed: the cone outlasts the oval under both nonzero rain rates, while the oval retains more sand in the dry comparison. The much shorter coupled lifetimes show how moisture evolution changes the size of the geometric advantage within these equations. This controlled model comparison does not measure a real beach's moisture effect. The fixed-C dry oval has not reached half volume by 48 h and is reported as a lower bound, with 78.99% of its initial sand still present.

Rain detachment is not always proportional to horizontal area alone. Defersha and colleagues found that soil erosion depends on runoff and initial wetness, and that its slope response need not be monotone [4]. Their soils and experimental rates do not calibrate beach sand. We therefore do not import their coefficients. Rills concentrating water down a foundation face, or saturation weakening a surface layer before its core, could change our geometric recommendation. A local rain-wash experiment is the most direct way to decide whether this additional structure is required.

10 Sensitivity identifies when to change the advice

We examined 18 deterministic combinations of three rain rates, three wave-direction offsets and two distance conventions. They are scenarios, not equally probable weather observations. The searched dry oval wins the aligned, toe-controlled dry comparison. At a 30° direction change, its dry lifetime becomes 7.16 hours while the circular cone remains at 9.00 hours. At 90°, the oval reaches half volume at 5.23 hours. At equal center distance, its longer seaward extent reduces the dry advantage sharply: 9.88 versus 9.05 hours.

In a further 24 deterministic stress cases, we crossed the two response tails, nominal peak fractions 0.005/0.010/0.020, low/high starting tide, and rain 0/5 mm h^{-1}. Table 6 exposes the resulting comparisons, with the larger lifetime in each pair emphasized. These are stress alternatives with no empirical probability or interval interpretation. Each design wins some cases, so neither dominates the complete set. The changing winners make calibration and useful-height preferences central to a recommendation.

The three retention thresholds give $T_{0.75} = 6.10$, $T_{0.50} = 15.62$ and $T_{0.25} = 21.90$ hours. These are distinct questions; a late surviving remnant cannot be substituted for a usable foundation. The reported ordering is monotone, as the erosion law requires.

Changing the common recession scale by factors 0.5, 1 and 2 produces lifetimes 28.69,

Table 6. Material stress and design ranking

Tail	θ_* (%)	Tide	Dry T_O/T_C (h)	5 mm/h T_O/T_C (h)
Exp.	0.5	Low	**15.62**/15.50	1.01/**3.75**
Exp.	0.5	High	**14.35**/11.04	0.90/**1.49**
Exp.	1.0	Low	**15.62**/9.00	1.08/**3.90**
Exp.	1.0	High	**14.30**/11.06	0.98/**1.99**
Exp.	2.0	Low	**15.62**/8.18	1.24/**4.36**
Exp.	2.0	High	**14.20**/10.95	1.13/**3.48**
Rational	0.5	Low	15.62/**15.74**	1.42/**4.99**
Rational	0.5	High	**14.37**/11.06	1.32/**4.46**
Rational	1.0	Low	**15.62**/9.44	1.58/**5.74**
Rational	1.0	High	**14.33**/11.06	1.48/**7.60**
Rational	2.0	Low	**15.62**/8.52	1.76/**6.91**
Rational	2.0	High	**14.24**/10.99	1.65/**10.02**

O: thin oval; C: circular cone. Larger lifetime is bold. Common $\theta_0 = 0.01$, aligned waves and equal toe distance; 24 deterministic comparisons, with no probability weights.

15.62 and 6.55 hours. These do not follow an exact inverse scaling because the tidal and moisture clocks are unchanged. In the constant-contact, fixed-moisture limit, (22) does give exact inverse scaling. This difference shows why a static surface-area ranking alone is insufficient for a rising tide.

The most valuable observations are the time and depth of first wave contact, prevailing and changing wave directions, a moisture-response curve for the actual packed sand, and the drying/drainage rate. Measuring v_w sets absolute hours, while those other inputs can change the chosen shape. No confidence interval or empirical winning probability is attached to an unmeasured parameter scenario.

11 Strategies beyond the initial geometry

Use position and tide timing. Equal-distance comparisons are necessary to isolate shape, but a builder can change location. In (4), $D \leq 0$ sets wave erosion to zero. Building farther inland or after the tide turns can therefore reduce wave exposure more than the modeled 73.51% aligned dry shape improvement in this example. Rain and drying remain active during that waiting time; local tide predictions and observed runup are needed to choose the position.

Compact uniformly and keep major loads low. Compaction supports a repeatable material comparison and is part of the preparation in the column study [1]. The relevant buckling scale for a column is $(GR^2/\rho g)^{1/3}$: greater stiffness helps, but increased density also increases weight. This column relation cannot simply certify a carved foundation. Broad, low, uniformly packed supports avoid relying on fragile isolated columns, and extra top loads should be checked separately.

Choose orientation only with reliable direction information. Aligning an oval's long axis with incoming waves reduces projected breadth; if directions change substantially, the cir-

cular footprint has a clear worst-direction advantage. The model gives an explicit cost for getting this wrong rather than assuming waves always strike the narrow side.

Manage water deliberately. Starting with already moist beach sand requires subtracting its existing water from any target addition. Rewetting a drying surface may help until it reaches the favorable material range; continued soaking can weaken it. In rain, allow drainage away from the base without cutting channels through supporting sand. A blanket instruction to "add more water" is incompatible with the nonmonotone material response.

Treat moats and repairs as separate resource decisions. A moat or sacrificial sand ridge can delay small swash, but a filled moat can concentrate flow at the base. The present model does not quantify that routing, so no universal moat benefit is claimed. Repairs add sand and labor; they can extend observed lifetime but violate a fixed-total-sand comparison unless their material budget is included. If B liters of maintenance sand are allowed, the sand ledger must include $V_0 + B$ rather than crediting geometry for unrecorded replenishment.

12 What is verified, and what remains to be measured

Numerical validation uses volume integration by horizontal slices, frontal-area quadrature, derivative and convexity checks for the analytic optimum, and direct-volume fixed-step RK4 instead of the model's shrinking-length adaptive RK45. All 26 named checks pass. The tested original, expanded-profile, response-tail and tide-phase lifetime differences are below 0.008 h; the dry RK4 result also agrees under step halving within 0.003 h. For the six fixed-resistance controls, the largest lifetime discrepancy is 0.0000746 h, with agreement on whether the half-volume event was reached. The grid sample sets and their regret tolerance are checked separately.

An additional author-external calculation, using physical V, W coordinates and DOP853, reproduced 98 retained coupled lifetime cases with a largest discrepancy below 1.2×10^{-4} h and a water-balance residual below 9.3×10^{-18} m^3. It independently checked the analytic geometry and feasible branches. These calculations validate the stated equations and numerical accuracy, not beach physics or a global dynamic search. Methods, identities, failed attempts and the distinction between author and external verification are documented in the reproduction appendix.

The main remaining scientific limits are identifiable. The shape closure cannot forecast undercutting, the wetness curve is not calibrated over all water regimes, direct rain erosion omits runoff routing, and the time-varying wave field is synthetic. Cohesive vertical walls, overhangs and profiles beyond the expanded search remain in the original problem's wider domain. The numerical bounds give useful constructible candidates but cannot exclude those alternatives. A full 3D simulation without data would add parameters rather than resolve these limitations.

13 Conclusions for the builder

A foundation's best geometry depends on the exposure direction, rainfall, moisture evolution, useful platform requirements and the survival metric. Our exact full-contact analysis gives a top/base radius ratio 0.366 when the rain/wave ratio is at most 0.386, and a circular boundary

optimum with a different top ratio above that threshold. In the illustrative tide-coupled comparison, the thinnest allowed curved oval has a conditional dry sand-retention advantage, while a compact circular cone is preferred when a raised platform, changed wave direction or rain matters. The dry optimum does not remain best in the two rain scenarios.

The model's static optimal mixture is near the nominal 1% bulk liquid fraction, with an explicit denominator and density-dependent mass conversion. A lifetime-optimal *initial* fraction can differ substantially because moisture changes before wave contact. Its near-best grid spans depend strongly on the response tail and tide phase; they are numerical tolerances rather than practical recipes. Fixed-resistance controls show that wave/rain geometry and moisture feedback both matter, and the stress comparisons show why no single design wins throughout the tested alternatives. Choose the site and tide phase first, compact the sand, keep supporting loads broad and low, and use drainage or maintenance only with their mechanisms and resource costs stated.

Fun in the Sun: Give your sandcastle a foundation that fits the weather

A little planning can matter more than an elaborate tower.

A tall sandcastle is a delight to build. Keeping it standing asks a different question: what happens when water reaches its base? Waves attack from the side, rain falls across the whole footprint, and the water inside the sand changes how well the grains hold together. A shape that does well against one of these forces may do badly against another.

We built a mathematical model to compare foundations made from the same amount of the same sand. It keeps a ledger of sand lost and water gained or lost. We call a foundation's lifetime the time until half its starting sand has gone. A castle's recognizable features could disappear earlier, so this measure is useful for comparing foundations, rather than promising that every window and turret will survive.

Start with a low, compact mound. In our sample conditions, a circular, gently sloping cone provided a useful raised base and handled changes in wave direction well. A wider oval could retain sand longer on a dry day when its long axis pointed straight toward the incoming waves. But the winning curved oval was only about 1.94 cm high, spread widely, and depended on waves arriving from the expected direction. That is a poor bargain if you want a raised castle platform.

Rain changes the bargain. A broad, thin foundation catches a lot of rain for each liter of sand it contains. Rain both washes grains away and changes the water stored among them. In our example, steady rain of 5 mm per hour made the dry-day oval lose half its sand much sooner. A compact circular cone lasted about 3.6 times as long under that same modeled rain. The numbers illustrate a comparison; they are not a countdown for your beach, whose sand and waves need local measurements.

A platform does not need straight vertical walls. Our simpler analysis found that a sloping oval mound with a small flat top can balance rain and wave exposure. When the modeled rain-to-wave erosion ratio was below its critical level, the top radius was about 37% of the base radius. Stronger relative rain selects a circular foundation with a different top ratio. This is a useful shape to try when you need a platform, but it is not a magic proportion for every weather forecast. The detailed model chose different shapes when moisture changed with time.

Check the tide before adding detail. If the waves have not yet reached the foundation, moving farther inland or choosing a better tide phase can buy more time than a small shape adjustment. Watch the largest swash that reaches the beach, and remember that the waterline can move. The foundation and the route water takes around it deserve attention before the decorative towers.

Use water carefully, and build for the conditions you see

Damp sand works because water can bind grains. Tiny bridges of water create an attraction between grains. More water is helpful only up to a point. Ordinary saturated sand loses the same liquid–air bridges that make a damp sculpture possible.

A laboratory study found the greatest stiffness near a reported liquid volume fraction of 1%. Our model uses a nominal water fraction of 1% of the compacted foundation's bulk volume when moisture stays constant. That means one liter of water stored in 100 liters of bulk foundation, including any water already in the sand. Water fills pores, so this is different from adding one liter to 99 liters of loose sand. It is also different from 1% by weight. The laboratory fraction and our bulk recipe are not a precisely established transfer to every beach.

Let a small test guide the mixture. Try equal, firmly packed samples of your actual beach sand with small measured additions of water. Compare how well they keep their shape and resist a repeatable gentle wash. Already wet sand may need little additional water. If the surface shines with free water or slumps, adding more is unlikely to rescue a tall feature.

Think about when the waves will arrive. A mixture that is strongest now may dry before the rising tide reaches it. Our drying model sometimes preferred a much wetter start as a water reservoir, with a broad range performing almost equally well. Different moisture-response curves gave different ranges, so this cannot become a universal recipe. Test how the actual sand changes during the expected waiting time, or carefully restore moisture before the surface becomes friable.

Pack evenly and keep supports broad. Compact small layers so that a weak pocket does not become the first place water opens a channel. A low broad support carries loads more reliably than a delicate tall column. Add the heaviest towers over the stronger central base and accept that carving deep into a support removes some of the sand that keeps it standing.

Send water away from the base. A shallow moat may intercept a small wash, but a moat that fills can guide water toward the very sand you want to protect. Avoid cutting drainage channels through a supporting face. We did not model the flow around moats, so there is no blanket promise that a deeper moat lasts longer.

Count repairs honestly. Extra sand and a little maintenance can extend the afternoon's fun. They are worthwhile choices, but the extra sand is part of the reason the castle lasts. A fair shape comparison must include it in the material budget.

The practical lesson is to build for the weather you can see: a compact mound for rain and uncertain wave directions, an aligned oval only when its low platform and directional dependence suit your plan, and enough water to keep the actual sand cohesive without soaking it. Give the tide and the supporting base the same care you give the towers.

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A Reproduction and evidence

Eight scientific attempts used the original workflow with Python 3.12.0, NumPy 2.5.3 and SciPy 1.18.1 in Praxis 0.7.1. The current run is `20261012T233556-3fd17927`. Its source/dependency snapshots, results, 26 checks and receipt are retained. Attempts 1/2 failed; 3–8 passed. Run 8 checks the immutable run 7 result digest, retains its numerical scenarios, and adds exactly six fixed-resistance solves and summaries of five retained initial-water grids. No geometry or mixture search was repeated. The frozen package includes the base run so that this dependency is reproducible.

The first run failed a neighboring-water comparison because a bounded scalar search missed a boundary, and dense interpolation introduced a small false volume rise during zero forcing. The corrected search compares endpoints and all sampled local peaks; the geometry search also compares domain corners; tightened integration controls remove the significant interpolation defect. The second run passed its mathematical checks but failed the pipeline's evidence-string contract. All failed runs and receipts remain retained. No failed search is treated as an infeasibility or global-optimality proof.

The first complete 17 page paper/run 3 and autonomous author initial final 19 page paper/run 6 remain immutable. The initial final PDF was compiled before feedback, then copied

unchanged after feedback arrived; the manifest preserves that distinction. Root cycle 01 identified missing long-axis feasibility in the constant-top-ratio claim and an incorrectly oriented wave arrow. Run 7 and the separately frozen cycle 01 paper corrected the circular boundary branch and diagram. Root cycle 02 requested scientific evidence tables, near-best water-grid ranges, the same-forcing fixed-resistance comparison and a nonempty-level/max-height sentence in the rain proof. The present cycle 02 version preserves that help rather than attributing it to the unassisted author phase.

The author validator is independently formulated but written by the same AI author. A separate AI reviewer used V, W coordinates and DOP853 on 98 retained coupled cases, checked geometry and feasible branches, and inspected the preceding complete paper. Its dated review files identify the fixed paper and evidence; it did not certify field parameters. The new six fixed-C solves are checked here by direct-volume RK4. Their auxiliary liquid-fraction overshoot at saturation is at most 6.4×10^{-7}, and the fixed resistance is independent of that auxiliary state. Event status and remaining volume were separately compared. Raw values were not clipped or rewritten after the run.

The model, independent validator and document generator accompany the numerical evidence. Figures use original vector geometry and all retained curve points. Source-reading ranges, retrieval failures, skill/lesson use and incidental exposure have separate records. Retrieval failures included a publisher redirect error, a repository anti-bot page, an unavailable optional Python import and a certificate error before successful retrieval of the original published text. Build/operational failures, including an article-specific template-check mismatch and the initial handoff JSON-list error, are retained in the case record. The validator does not import the model.

The historical 20 page solution and 24 page combined limit cover the summary, contents and two-page article as specified; references and appendices follow and are excluded. The separately paginated AI disclosure records this present-day exercise, without claiming a 2020 formal submission. Exact page counts and full-page reviews bind the PDF.

Report on Use of AI

An OpenAI Codex AI author performed the original-task reading, model design, source reading, Python implementation, workflow runs, manuscript and article writing, vector-figure generation and author numerical checks within a fixed 240 minute window. The actual model identifier, token total and monetary cost are unavailable and are not invented. No human field experiment or independent human calculation is claimed.

The external material consisted of original-domain research, an official NOAA educational page and an original-author mathematical survey. Actual reading ranges and failed requests are recorded. The original 2020 B statement came from a fixed third-party COMAP mirror; its byte identity with an official download is unverified. Laboratory sand/water results are not represented as calibrated beach observations.

A search for the 2012 material paper's PDF accidentally returned a snippet from a third-party 2020 B student document, showing a frustum/8:1 suggestion and general tips. That source was never opened or downloaded and was stopped. Generic frustum geometry was considered earlier, but this does not prove the snippet had no influence. The exercise is not called unexposed or blind. Its numerical ratios and proof do not use that snippet as evidence.

The FIRST COMPLETE, author initial final and feedback cycle 01 artifacts are immutable. Root cycle 01 identified missing long-axis feasibility and a wave-direction error. Root cycle 02 requested evidence presentation, near-best initial-water ranges, a small fixed-resistance control and explicit nonempty levels in the broad rain proof. The author implemented and checked these separately attributed revisions within the original eight-attempt/240-minute window; no additional beach experiment or field calibration was performed. Author calculations and external AI review remain distinct, and no award label is treated as a proof. Exact times, feedback records and hashes are preserved. Template team number 0000000 is an unassigned exercise identifier, not an official control number.